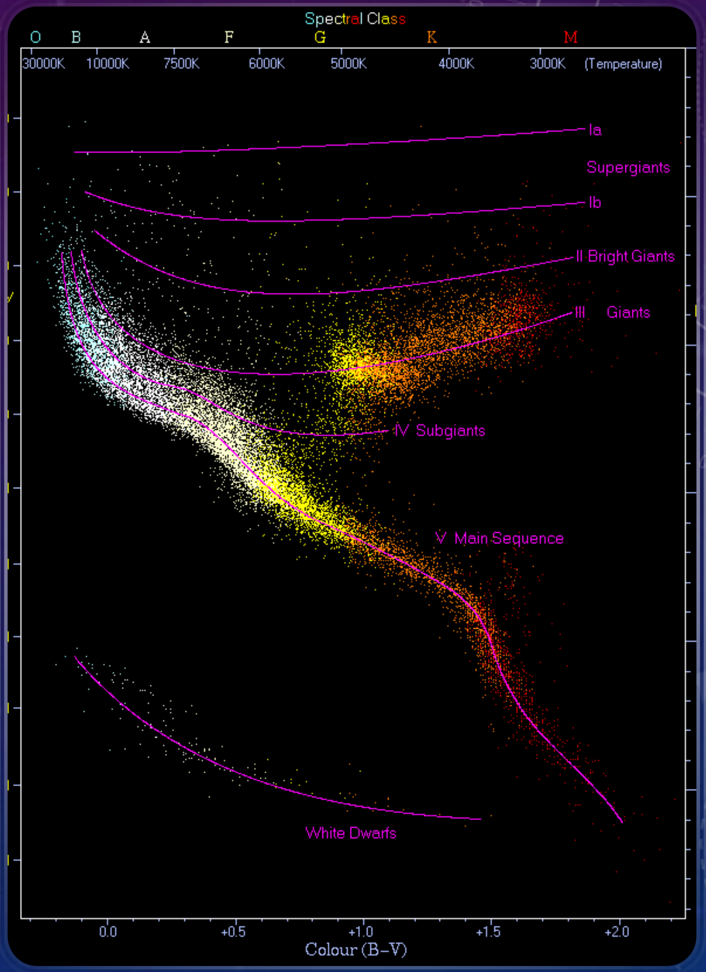
HISTORY

1920s: Hertzsprung-Russell Revolution
Ejnar Hertzsprung and Henry Norris Russell independently discovered the correlation between luminosity and temperature, creating the HR diagram - the foundation of stellar evolution theory.[1]

1930s-1950s: Stellar Structure Breakthroughs
Arthur Eddington established mass-luminosity relations; Subrahmanyan Chandrasekhar defined white dwarf mass limits; Fred Hoyle formulated nucleosynthesis theory.



DATA COLLECTION

- **Source**: Gaia DR3 (ESA's space observatory).
- **Filters**:
    - Stars within **100 pc** (parallax > 10 mas).
    - High-quality astrometry (ruwe < 1.4).
    - Bright stars (phot_g_mean_mag < 15).
- **Sample Size**: 100,000 stars.
- **Key Attributes**:
    - Parallax, BP/RP magnitudes, temperature (teff_gspphot).
    - Variability flag, radial velocity.

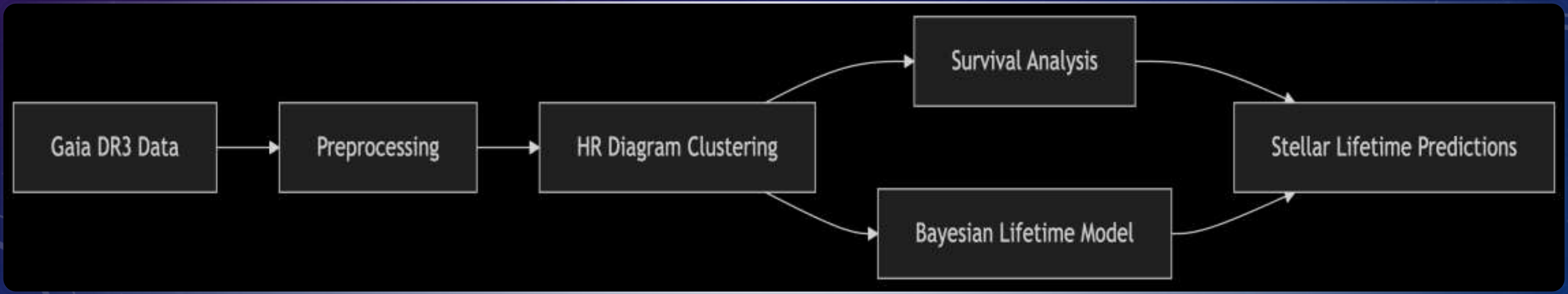
```
query = """
SELECT TOP 100000  -- Start smaller
    ra, dec, parallax, parallax_error,
    phot_g_mean_mag, phot_bp_mean_mag,
    phot_rp_mean_mag,
    bp_rp, radial_velocity, ruwe,
    teff_gspphot, phot_variable_flag
FROM gaiadr3.gaia_source
WHERE
    parallax > 10
    AND bp_rp IS NOT NULL
    AND phot_g_mean_mag < 15
    AND ruwe < 1.4
    AND teff_gspphot IS NOT NULL
"""
```

PYTHON LIBRARIES USED

Data Query	astroquery.gaia
Data Processing	pandas, numpy
Visualization	matplotlib, seaborn
Clustering	scikit-learn (KMeans, GMM)

PLANNED WORKFLOW

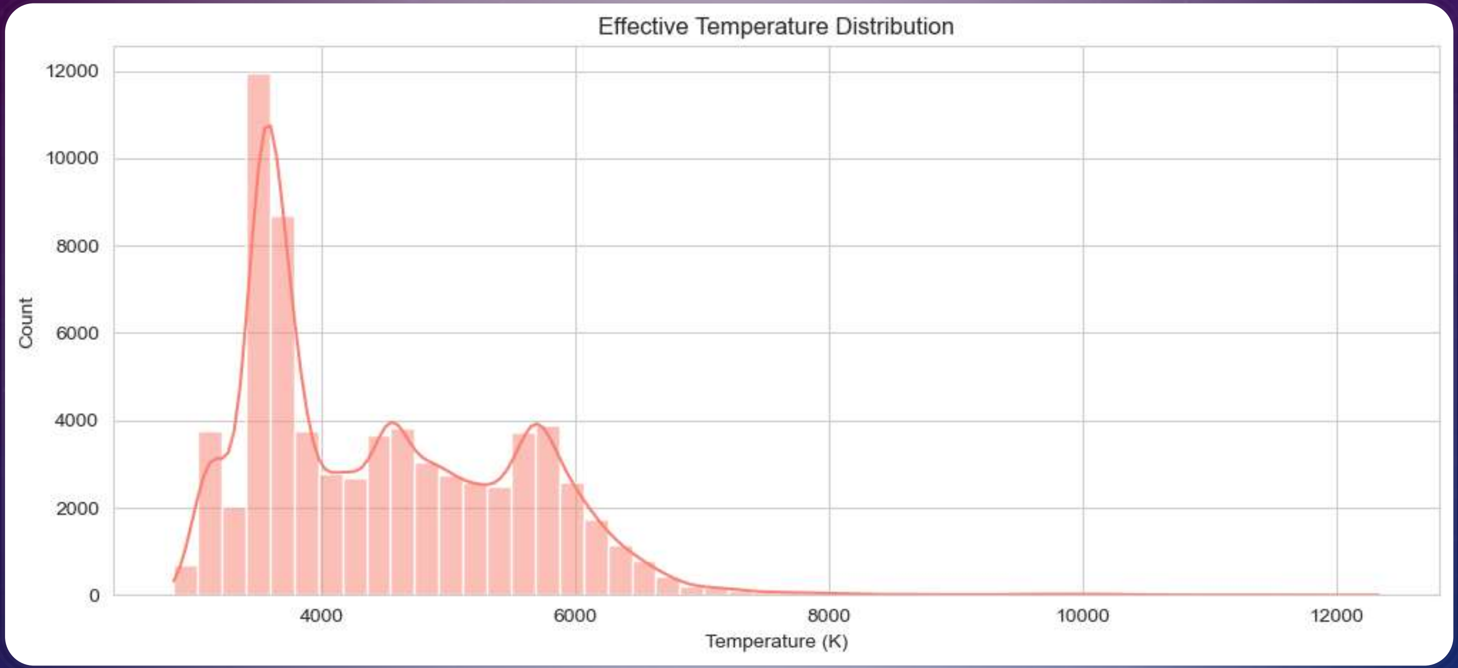
- Data preprocessing for the parallax and absolute magnitude
- HR diagram clustering on the threshold of bp_rp and abs_mag, with ML models like GMM, DBSCAN.
- Stellar lifeline prediction using the models like Kaplan-Meier estimator and Cox PH.



RESULTS SO FAR

- Initial Shape: (73,148 rows, 12 columns) -> Final Shape: (70118, 12)
- Filter Applied:
  1. Keep stars with parallax error < 20%
  2. distance_pc= ${}^{1000}/_{parallax}$
  3. M_G = phot_g_mean_mag−5 x log(distance_pc)+5

	parallax	distance_pc	phot_g_mean_mag	abs_mag
0	13.398257	74.636572	13.470652	9.105894
1	19.588654	51.049959	6.224640	2.684663
2	10.326266	96.840427	13.695807	8.765524
3	11.136129	89.797807	14.756236	9.989907
4	17.357228	57.612888	10.476051	6.673453

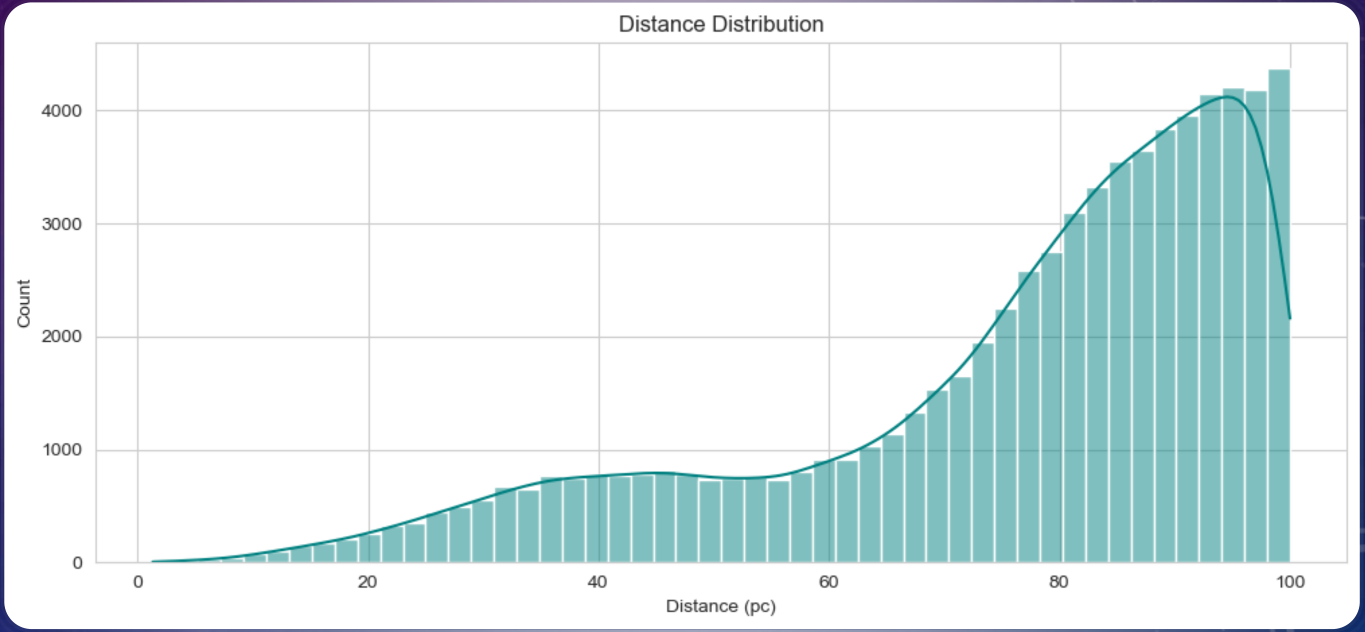


EFFECTIVE TEMPERATURE DISTRIBUTION

- The distribution is skewed , with a peak around 3,500–4,000 K .
- Most stars have temperatures in the range of 3,000–6,500 K .
- Few stars are present at higher temperatures (>6,000 K).
- The peak suggests that main-sequence stars dominate the dataset.

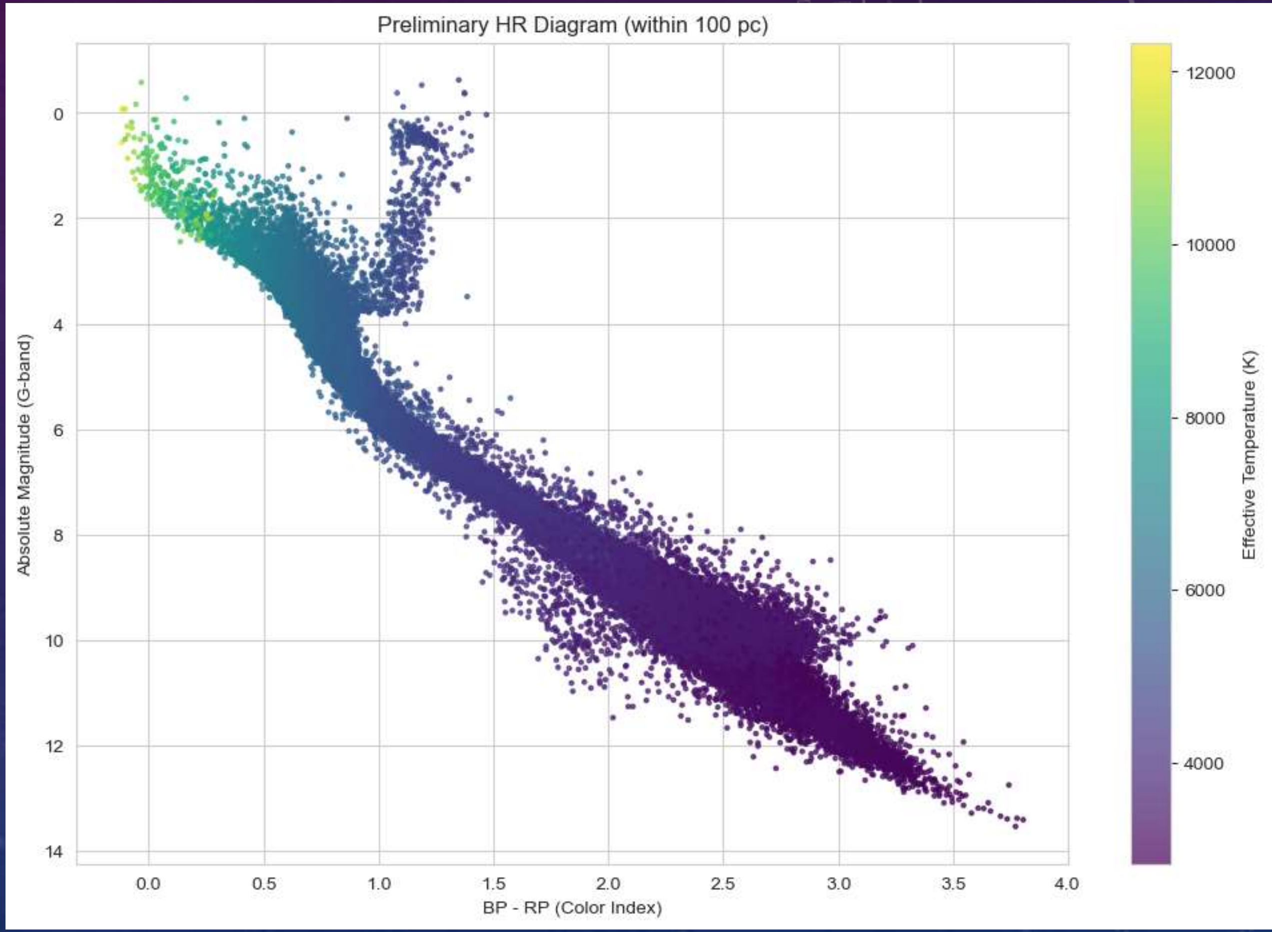
DENSITY DISTRIBUTION

- The distribution shows a peak around 85–95 parsecs .
- There is a gradual increase in the number of stars as distance increases up to ~95 pc.
- Beyond 95 pc, the count decreases sharply.
- Most stars in the dataset are located within 65–95 parsecs , indicating a focus on nearby stars.
- The decline beyond 100 pc suggests that the dataset prioritizes stars with closer stars.



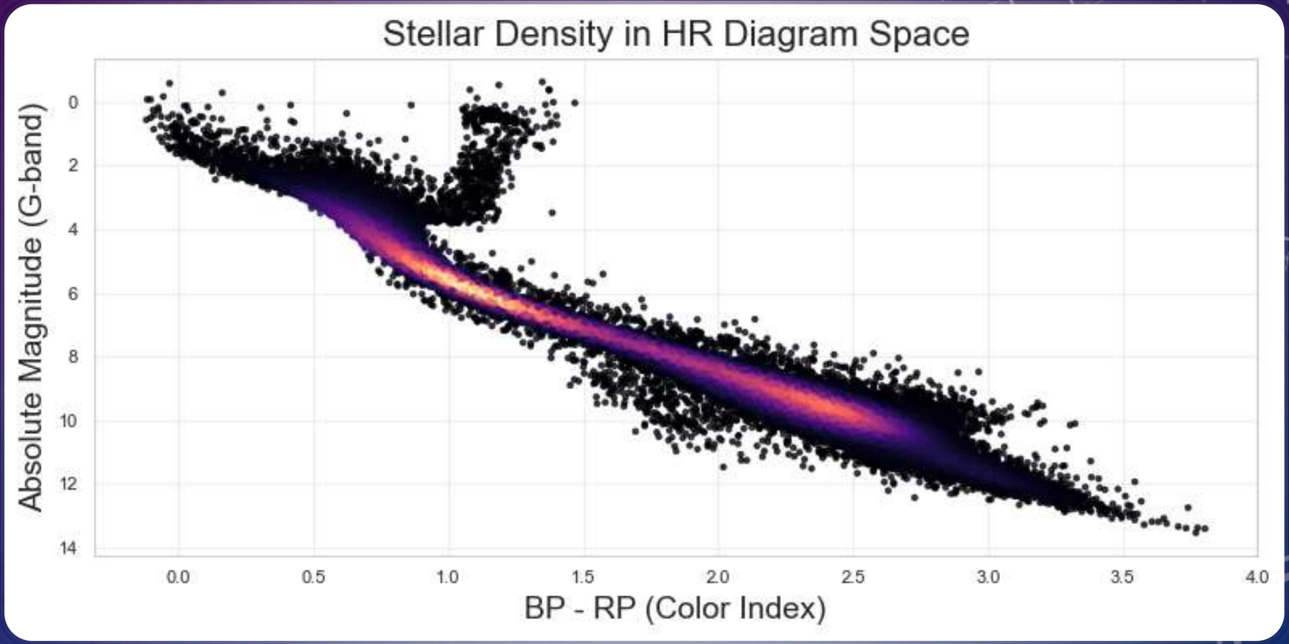
PRELIMINARY HR DIAGRAM

- A clear diagonal band where most stars lie, indicating a strong correlation between temperature and luminosity.
- Cooler stars (higher BP - RP) have lower temperatures and are darker, which can be red giants.
- Hotter stars (lower BP - RP) have higher temperatures and are brighter, can be white Dwarfs.



STELLAR DENSITY

- The diagonal band is clearly visible, representing the main sequence where most stars lie.
- Brighter Regions (Red/Orange): Indicate areas with higher stellar density.
- Darker Regions (Black): Indicate areas with lower stellar density.
- The main sequence shows a high density of stars, as expected.
- The main sequence is the most densely populated region, reflecting the prevalence of active hydrogen-burning stars.



REFERENCES

[1] WIKIPEDIA CONTRIBUTORS. (2025, APRIL 23). HERTZSPRUNG–RUSSELL DIAGRAM. IN *WIKIPEDIA, THE FREE ENCYCLOPEDIA*. RETRIEVED 12:57, JUNE 13, 2025,FROM HTTPS://EN.WIKIPEDIA.ORG/W/INDEX.PHP?TITLE=HERTZSPRUNG%E2%80%93RUSSELL_DIAGRAM&OLDID=1287087896.

[2] MALTSEV, K., SCHNEIDER, F. R. N., RÖPKE, F. K., JORDAN, A. I., QADIR, G. A., KERZENDORF, W. E., RIEDMILLER, K., & VAN DER SMAGT, P. (2023). SCALABLE STELLAR EVOLUTION FORECASTING. *ASTRONOMY AND ASTROPHYSICS*, *681*, A86. HTTPS://DOI.ORG/10.1051/0004-6361/202347118.

[3] KAPLAN, E. L., & MEIER, P. (1958). NONPARAMETRIC ESTIMATION FROM INCOMPLETE OBSERVATIONS. *JOURNAL OF THE AMERICAN STATISTICAL ASSOCIATION*, *53*(282), 457–481. HTTPS://DOI.ORG/10.1080/01621459.1958.10501452.